→ Rear axle mount chassis Analysis.

Types of analysis.

- Torsional test
- Structural Analysis (Bending)
  - static load
  - Dynamic Bump.
- Lateral load Transfer.
- Braking & Acceleration

(i) Torsion Test. ——→ [EXPLICIT DYNAMICS]

Material for rear axle ⇒ AISI 4130 [chromyl steel] [Good strength to wt ratio & excellent fatigue resistance under cyclic loads.]

$\rho = 7850 \text{ kg/m}^3$

Young's E = 205 GPa

Poisson $\gamma = 0.29$

Tensile yield strength = 460 MPa.

Tensile ultimate strength = 560 MPa.

Kart + driver mass = 160 kg.

Rear weight Bias = 60%.

$= 160 \times 9.01 \times 0.6$

$= 941.76 \text{ N}$

Tyre radius = 5.5 inch = 0.139 m.

μ(tyre) = 1

$T_{engine\ max} = 12 \text{ N-m}$ (125 cc)

gear = (5)

chain $\eta = (0.92)$

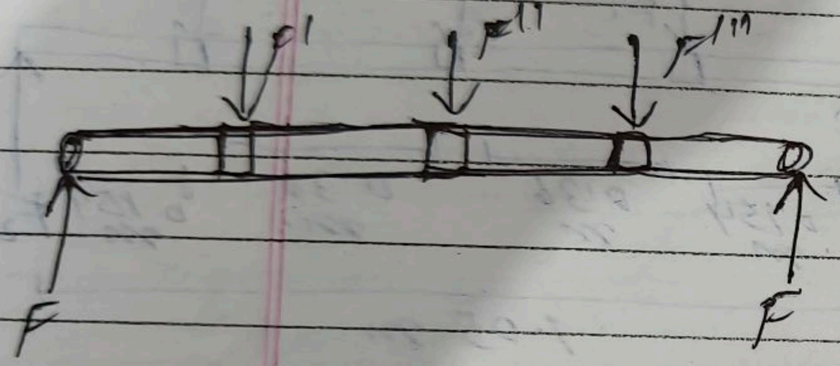
$T_{drive} = 12 \times 5 \times 0.92 = (55.2 \text{ N-m})$

b) [for performing this test I locked both wheels & provided max. T that can be transferred from chain to axle at chain mount.]

(ii) Structural Analysis (Bending)

↓

Static load
(due to its own weight.



$\Sigma F = (F' + F'' + F''')$

↓

Total weight of go-kart on back. ⇒ 60% of 150

⇒ (90 kg)

Let equal distribution of weight

$F' = F'' = F'''$

↳ Let engine & all are distributed evenly.

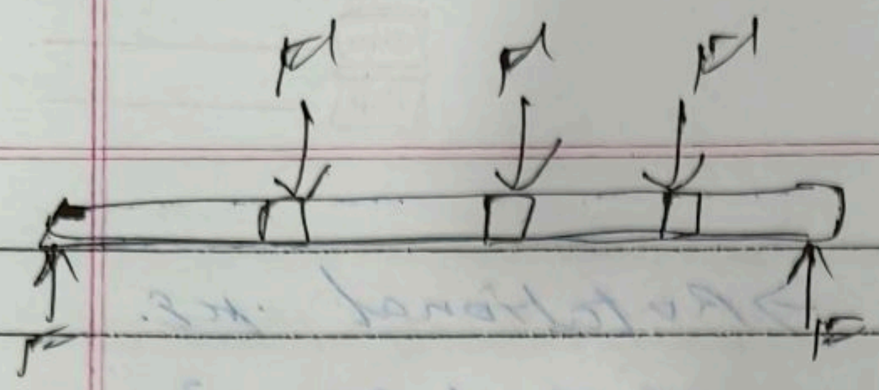
(iii) ~~Dynamic~~ ~~Bump~~ / Acceleration

[Testing the rear axle when the kart moves ~~over a~~ from ~~sudden bump~~ & acceleration.

In this everything is same as static load just the value of F' & F will increase

[REST SAME]

(i) Torsi
Mar
$\ell = 7$
Young
Poisson
Tensile
yield
streng
tensile
ultim
streng

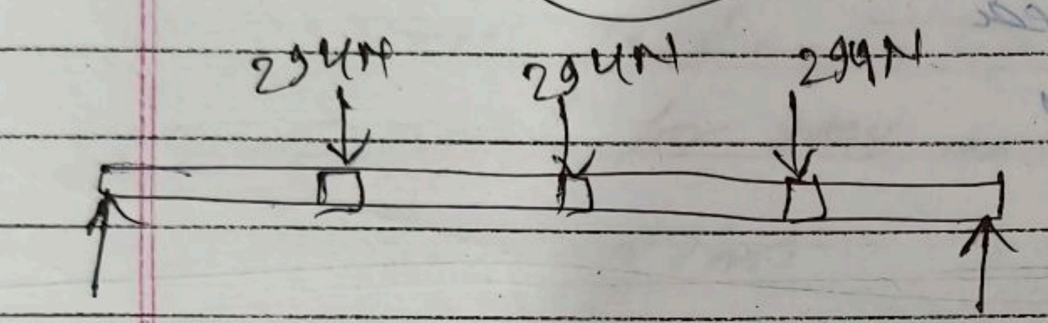


$3F' = 90 \times 9.8\ N$

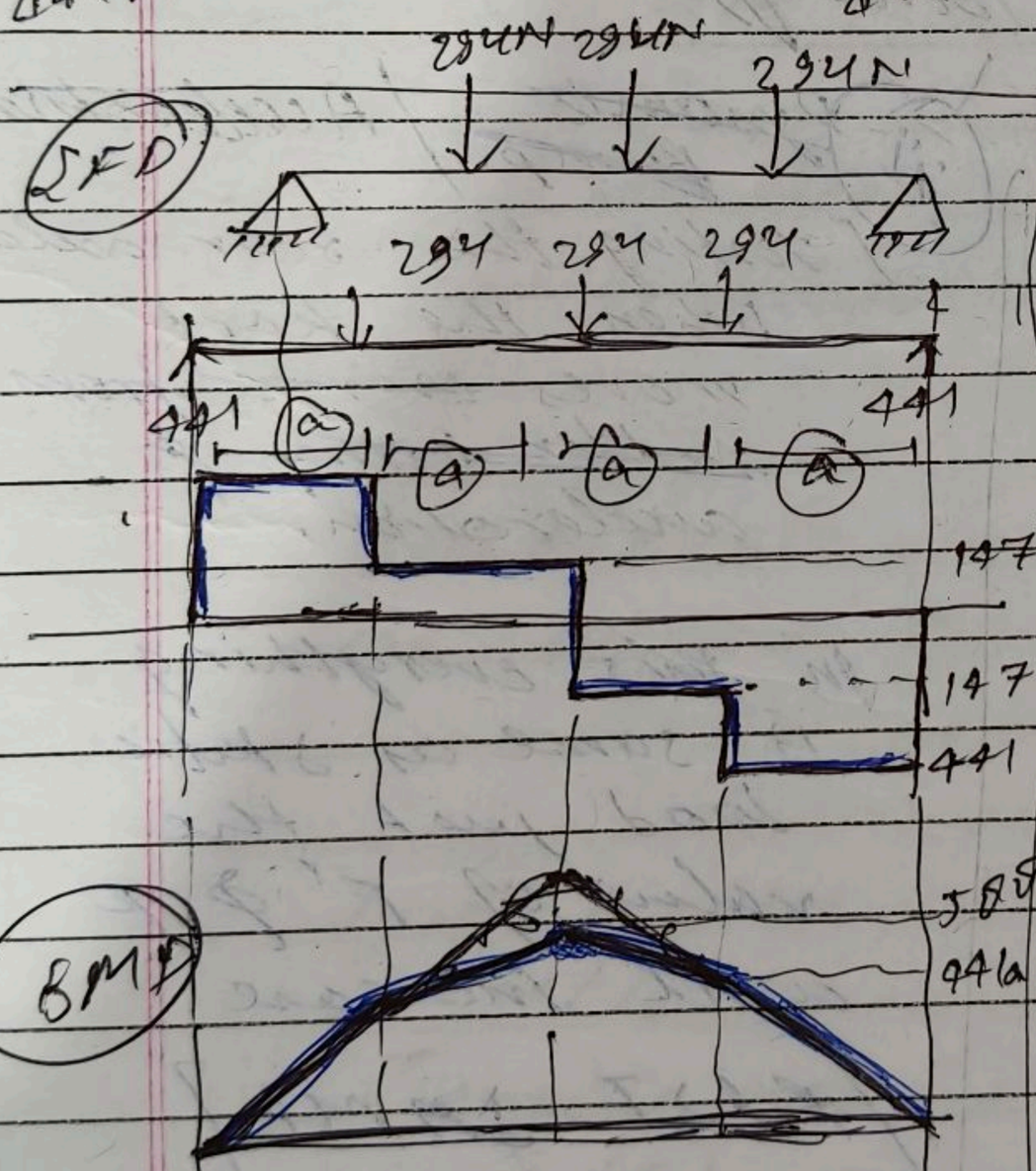
$F' = 294\ N$

$2F = 90 \times 9.8\ N$

$F = 441\ N$



SFD

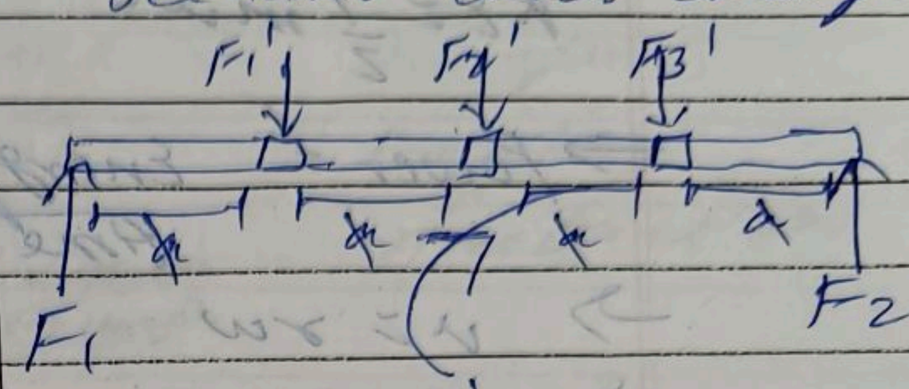


BMD

Validated by Total deformation in ANSYS as central is most bend & sides are least.

(iv) Lateral load Transfer.

On cornering the scenario will change.



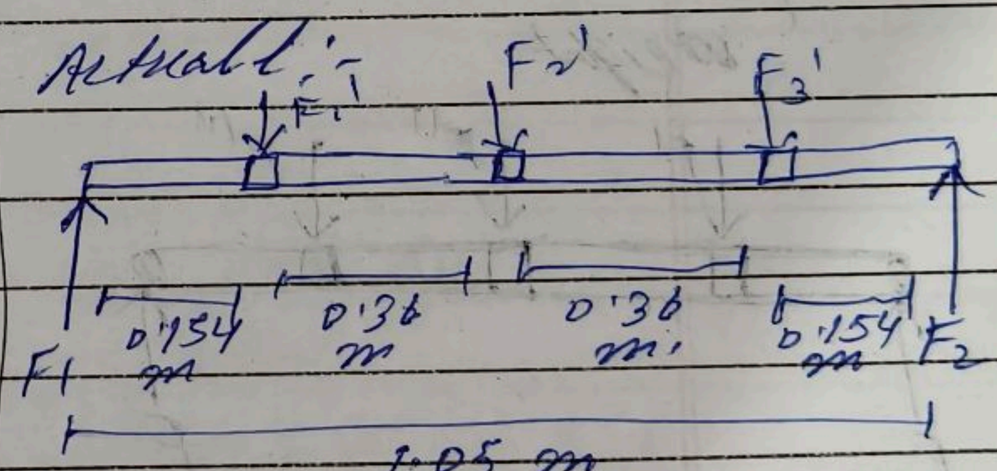
Kart has taken a right turn

[So wt will transfer on left side].

Let $v = 40$ kmph $= 11.11$ m/s.

wt. on axle when kart is on rest $\approx 90 \times 9.8 = 882\ N$

Actual :-



$F_1 + F_2 = F_1' + F_2' + F_3'$

Let :- at Limit

Lateral acc. ≈ 1.5 g.

Track width $= 1.25$ m

~~COM height = 15 in = 0.38 m~~

$F_{lateral} = m_{veh} \times a_y$

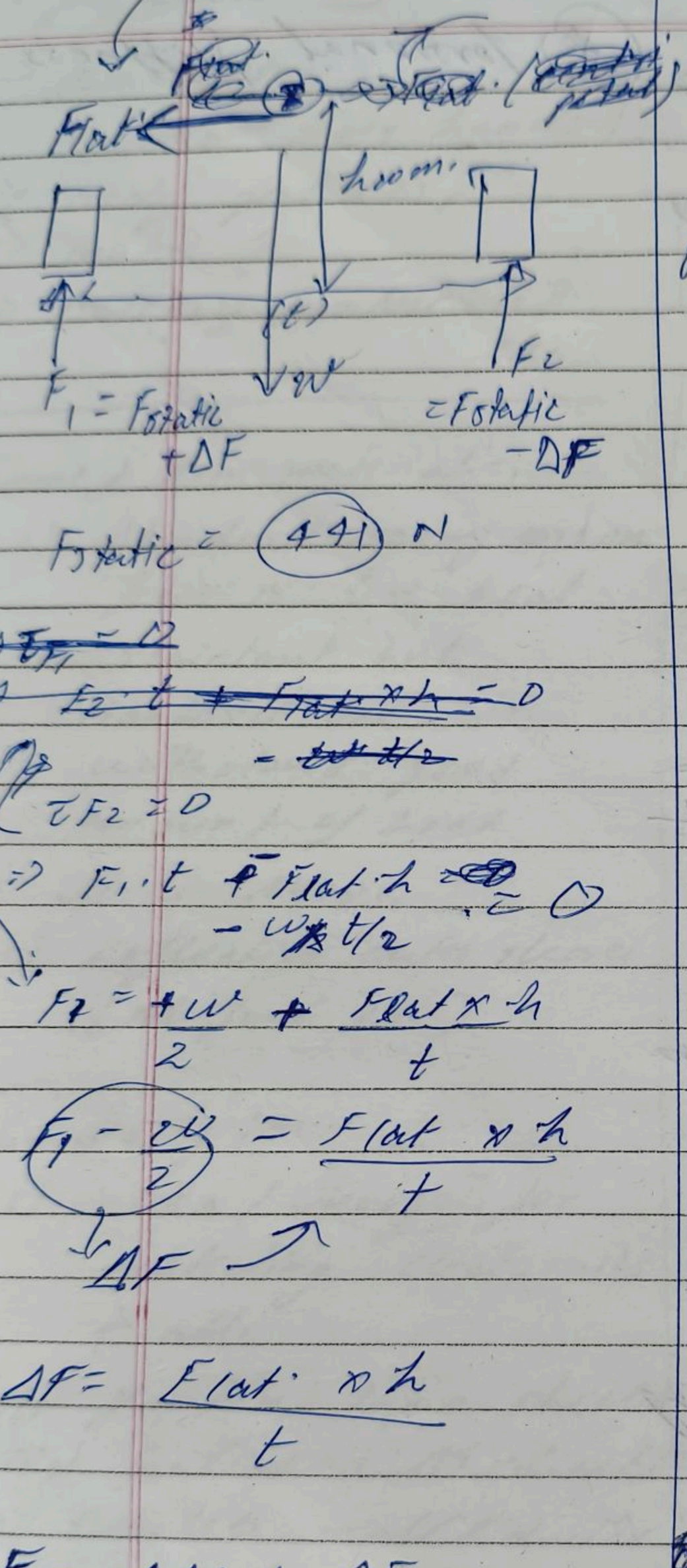
$= 90 \times 1.5 \times 9.8$

$= 1323\ N$

COM height ≈ 15 in.

$= 0.38$ m

weight transfer = $\frac{F_{lat} \times h_{com}}{\text{track width } (t)}$



$F_{static} = 441$ N

~~$\Sigma F_1 = 0$~~

~~$\Rightarrow F_2 \cdot t + F_{lat} \times h = 0$~~

~~$= W \cdot t/2$~~

$\Sigma F_2 = 0$

$\Rightarrow F_1 \cdot t + F_{lat} \cdot h - W \cdot t/2 = 0$

$F_1 = \frac{W}{2} + \frac{F_{lat} \times h}{t}$

$F_1 - \frac{W}{2} = \frac{F_{lat} \times h}{t}$

ΔF

$\Delta F = \frac{F_{lat} \times h}{t}$

$F_1 = 441 + \Delta F$

$= 441 + \frac{1323 \times 0.38}{1.25}$

$= 441 + 402.1$

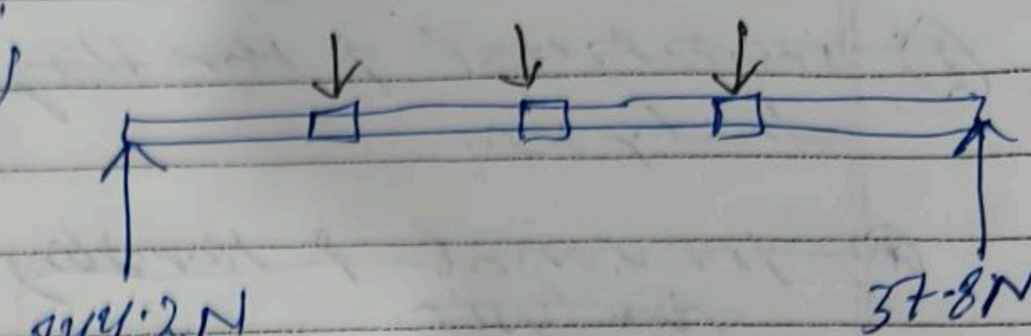
$= 843.1$ N

$F_2 = 441 - 402.1$

$F_2 = 37.8$ N

20.

Date
Page

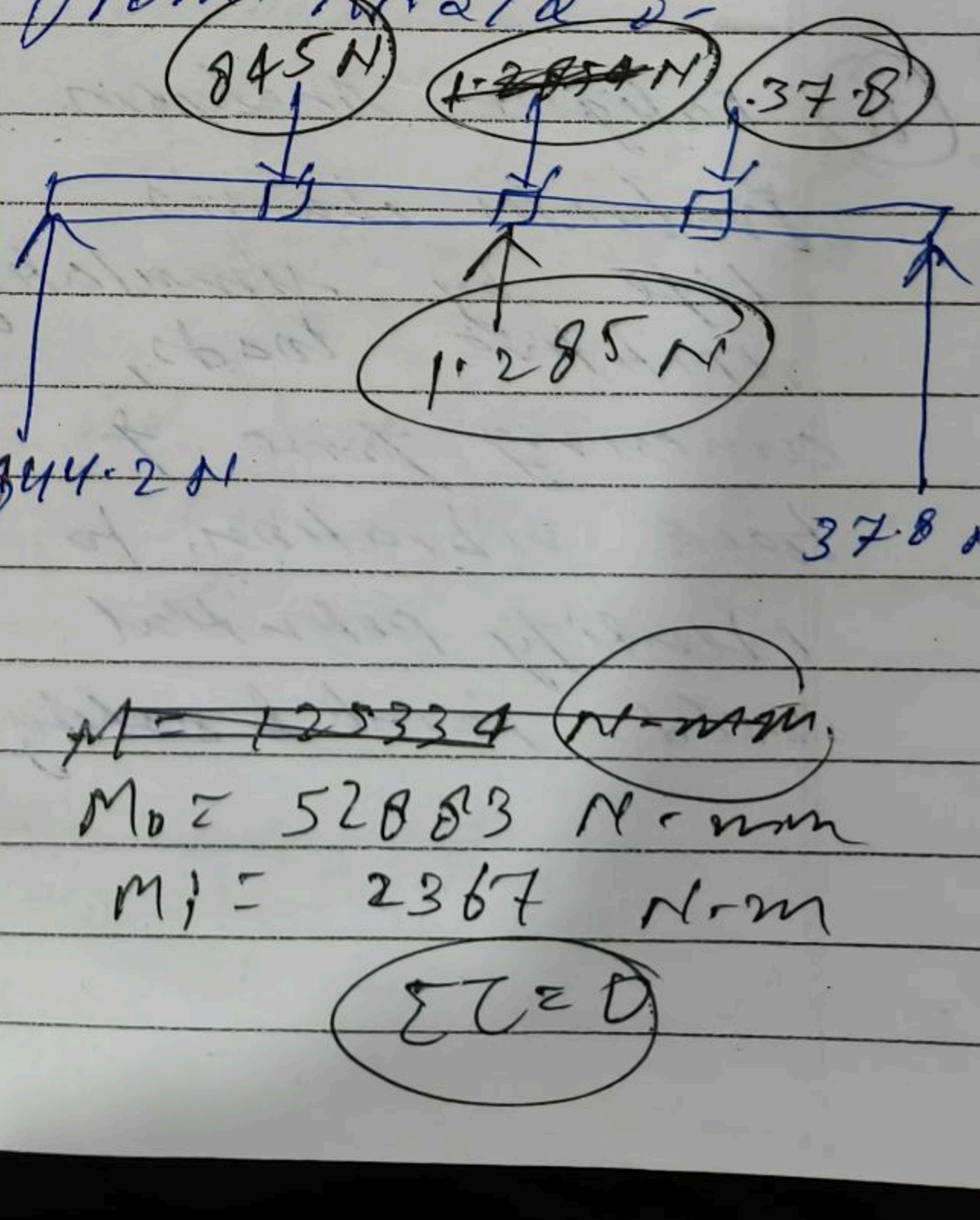


↳ [I will run the analysis at this point & will observe the ron. at bearing in analysis]

↳ Also there is no suspension in go-kart

So. [No need to worry about Roll much.]

from ANSYS :-



~~$M = 125334$~~ N-mm

$M_o = 52883$ N-mm

$M_i = 2367$ N-m

$\Sigma T = 0$